

ClimateMetrics v1.1

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Frequently Asked Questions

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This document compiles questions received from clients, webinar attendees, and prospects, together with detailed answers on ClimateMetrics' methodology, platform, and commercial offering.

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1. General Methodology

Q What climate scenarios and time horizons are available?

For physical risk, the platform provides projections under two IPCC scenarios: SSP2-4.5 (moderate emissions trajectory) and SSP5-8.5 (high emissions, business-as-usual trajectory). Physical risk is projected annually from 2025 to 2100. Historical data is also available for baseline comparisons.

For transition risk, eight scenarios are available from the Scientific Climate Scenarios framework, covering a range of decarbonisation ambitions. Scientific Climate Scenarios are an advanced climate scenario framework developed by the EDHEC Climate Institute to support forward-looking climate risk analysis, valuation, and investment decision-making. Building on existing NGFS climate scenarios, the framework enhances their usability for financial institutions and investors through three major innovations: granularisation, extension, and probabilisation of climate scenarios. The framework covers the 6 main NGFS scenarios, among which Current Policies and Net Zero, while introducing two additional scenarios - Climate Destabilisation and Climate Breakdown - with additional higher-warming pathways designed to better capture tail climate risks and potential climate destabilisation dynamics

Transition risk is projected to 2060 in 5-year steps. The default rating scenario is the Expected scenario.

Q How does the platform handle shorter-horizon climate impacts, given that SSP scenarios are typically relevant at 10+ year timescales?

That is a fair point. Our platform covers time horizons from the present day to 2100 for physical risk and to 2050 for transition risk, so near-term impacts (e.g. 2025-2035) are included within the output time series.

For physical risk, we use SSP/RCP-based scenarios (SSP2-4.5 and SSP5-8.5). While these are long-term frameworks, the near-term portion of the time series reflects the current hazard environment with modest scenario divergence - appropriate for strategic risk assessment over 5-15 year horizons.

For transition risk, we use Oxford Economics scenarios calibrated to shorter-term policy cycles, which provide meaningful differentiation over 5-10 year horizons. Very short-term stress-testing applications (intra-year or 1-3 year horizons) are outside the current platform scope, which is designed for strategic, medium-to-long-term investment decision-making.

Q How often are the models updated?

Our underlying models and reference distribution are updated annually, ensuring that ratings reflect the latest available climate science, emissions datasets, and damage function calibrations. Incremental patch releases are also deployed throughout the year as methodology improvements are validated by the Methodology Committee.

2. Data Inputs & Asset Coverage

Q How does the platform handle geolocation? What input types are supported?

At minimum, the GPS coordinates (longitude/latitude, WGS84) of the asset centroid are required. Additionally, users can define:

- A point with a buffer zone (radius in metres around the centroid).
- A line (e.g. a pipeline or road) with or without a buffer zone.
- A polygon (GeoJSON) representing the full asset footprint.

Using a polygon or precise boundary significantly improves the accuracy of the physical risk assessment by capturing the actual spatial extent of the asset, rather than assuming a circular buffer.

Q If you only have an address and not coordinates, can the platform still find the coordinates?

Yes. If you input a street address (with postal code), the platform will automatically geocode it to GPS coordinates (latitude/longitude) using an address geocoding service. Coordinates are then used as the basis for all physical hazard extractions. Providing coordinates directly is preferred for precision, but a well-formed address is fully sufficient to run an assessment.

Q How do you deal with linear or geographically dispersed assets (e.g. pipelines, roads, rail lines)?

Linear assets are supported natively. Users can define an asset using a line geometry (with or without a buffer), and the platform computes hazard exposure along the full length of the asset's footprint. For long assets crossing multiple geographies or risk zones, the platform aggregates hazard intensities across all relevant data pixels using zonal statistics.

Where a dispersed asset contains distinct sub-assets with meaningfully different risk profiles (e.g. a wind farm with turbines across multiple sites), it is recommended to assess each sub-asset separately and then aggregate at portfolio level.

Q If there are multiple distinct asset types within a single polygon, should each be assessed as a separate run?

Yes. In ClimateMetrics, one request equals one asset analysis. Each asset must be uploaded individually with its own asset type, geolocation, and financial data, since the asset type drives the damage functions, emissions model, resilience strategies, and sector parameters. Portfolio-level aggregation then allows combining them for a consolidated view, with each portfolio aggregation counting as one additional request.

Q What asset types are covered?

The platform covers more than 130 asset types, classified according to the Infrastructure Company Classification Standard (TICCS) developed by Scientific Infra & Private Assets. These span all major infrastructure sectors: energy (power generation, transmission, distribution), transportation (roads, airports, ports, rail), water (treatment, distribution, wastewater), digital infrastructure, social infrastructure, and more. Residential buildings are also covered.

Q Can you provide examples of asset types for which damage curves are available?

Flood damage functions are available for a wide range of infrastructure asset types, including roads, power generation plants, wind farms, gas water mains, water tanks, water treatment facilities, and oil refineries. Storm (wind) damage functions are calibrated to infrastructure building typologies across countries. Wildfire damage functions are also implemented at the asset level.

In total, the platform encompasses over 40 damage functions across hazards, asset types, and geographies. Where a dedicated function is not available for a specific country or sub-sector, country proxies are applied based on the closest comparable calibrated dataset.

Q Does ClimateMetrics cover agriculture-related assets?

ClimaTECH is currently focused on infrastructure assets, which is the primary asset class covered by ClimateMetrics. Agriculture-related assets and commodity production are not yet fully supported within the ClimaTECH framework. However, the extension to additional sectors - including agriculture - is actively being developed.

In the meantime, farm assets can be assessed for physical climate risk exposure (flood, wildfire, heat, storm) using the closest available asset type mapping, though transition risk parameters would be less precise. We welcome enquiries from users with agricultural exposure who want to explore what is possible today.

Q Can ClimateMetrics produce a climate rating for a specific private home - for example, a residential property in London?

Yes, in principle. Any asset can be assessed based on GPS coordinates or a polygon footprint, and residential buildings are among the supported asset types. A mortgage lender or investor could upload a residential address, assign the appropriate asset type, and receive a physical risk rating (flood, storm, wildfire, heat) for that specific location. This is directly applicable to mortgage book climate stress-testing.

One nuance: the transition risk module is oriented toward infrastructure and corporate assets (carbon intensity x carbon price), so it would be less meaningful for a private dwelling in isolation. The physical risk output, however, is fully applicable and can be run for large volumes of residential addresses via bulk Excel import.

Q Do users need to upload their own assets? Does the platform provide pre-loaded issuer or asset data?

Users upload their own assets. The platform does not come pre-loaded with issuer or third-party asset data. Users provide geolocation, asset type, and financial information (asset value, revenue, and optionally Scope 1 and 2 emissions) either manually through the platform interface or via bulk Excel import. Once uploaded, the climate assessment runs automatically.

Q Do you use financial statements for quantification or resilience testing?

Financial statements are not directly ingested, but two financial inputs drive the quantification: asset value (used to express physical damage as a percentage of replacement value) and asset revenue (used for transition risk and heat stress financial impact). Users typically derive these from their own internal records or published financial statements.

If these values are not provided, the platform applies sector-level default ratios. Reported Scope 1 and 2 emissions from sustainability reports can also be uploaded to replace model-estimated emissions intensities, significantly improving transition risk precision.

3. Physical Risk

Q What physical hazards are covered?

The platform currently covers eight physical hazards: inland flooding, coastal flooding, extratropical cyclones (storms), tropical cyclones (storms), wildfire, heat stress, precipitation, and drought. For each hazard, multiple indicators are available - more than 30 in total - assessed at three return periods (typically 20-year, 100-year, and 1,000-year events) for acute hazards. Water stress and additional chronic indicators are also available.

Future developments include the integration of earthquake, subsidence, and sea-level rise indicators, along with additional scenario coverage (SSP1-2.6).

Q What are the resolutions of your hazard maps, and is downscaling involved?

Resolution varies by hazard:

- Floods: 30m x 30m - the highest resolution available in any commercial climate risk platform.
- Extratropical storms: ~28km x 28km.
- Tropical cyclones: ~11km x 11km.
- Wildfires: 500m x 500m
- Heat stress (WBGT): ~25km x 25km.

Downscaling is involved for several hazards.

Q Does ClimateMetrics cover water runoff from heavy rainfall (pluvial flooding)?

Yes. Flood risk is assessed across three distinct sub-hazards:

- riverine (fluvial) flooding
- surface water (pluvial) flooding
- coastal flooding

The second directly captures the runoff dynamics associated with intense rainfall events. Both contribute to the overall flood damage time series projected from 2025 to 2100.

Q What are the sources of the underlying hazard data?

The platform draws on a range of high-quality geospatial and scientific datasets:

- Floods: Fathom Global Flood Map
- Extratropical cyclones: ERA5 reanalysis wind data from ECMWF; CORDEX regional climate model ensembles for future scaling.
- Tropical cyclones: STORM synthetic track model (Bloemendaal et al.).
- Wildfires: Fire Weather Index (FWI) based on meteorological projections.
- Heat stress: NASA NEX-GDDP-CMIP6 (WBGT computed from downscaled temperature and humidity).
- Water stress: WRI Aqueduct 4.0.

The platform does not use satellite imagery directly - the underlying datasets are derived from reanalysis, climate model output, and remote sensing observations, rather than raw satellite feeds.

Q Is the entire world covered, or only Western Europe?

Coverage is global for all primary hazard modules. Inland and coastal flooding, tropical and extratropical cyclones, wildfire, and heat stress all have global land coverage. Transition risk coverage follows the availability of country-level revenue, emissions, and carbon price datasets across 27+ countries.

A few technical caveats apply: tropical cyclone projections run to 2060 rather than 2100; some damage function country proxies are applied in regions without directly calibrated curves; and offshore assets receive zero scores for flood and wildfire risk by definition.

Q Does ClimateMetrics account for cascade failure or infrastructure dependencies?

The current model assesses each asset independently based on its own geographic footprint, asset type, and declared characteristics. Infrastructure interdependencies - such as a power plant losing output because a grid substation has failed, or a water treatment facility becoming inoperative due to road network disruption - are not currently modelled.

This is a recognised limitation and a relevant avenue for future development. For assets with strong operational dependencies on other infrastructure, we recommend supplementing the platform's output with an operational risk assessment.

Q Does physical impact include only direct damage, or does it also capture business interruption?

Both are captured. ClimateMetrics quantifies two distinct financial impacts for physical hazards:

- Damage: the cost of physical reconstruction, expressed as a percentage of asset replacement value. This impacts CAPEX.
- Disruption: service or operational interruption caused by the hazard, expressed as a percentage impact on asset revenue. This impacts EBITDA and cash flows.

Disruption is computed using TICCS sector-specific disruption factors that convert physical damage fractions into expected service interruption periods. Both metrics are available in the platform outputs and reported as separate line items.

Q Is there any hail-related analysis available?

Hail is not currently included as a separate hazard in ClimateMetrics. It has been identified as a candidate for a future version but faces data availability and modelling feasibility challenges at global scale.

We currently cover the four most financially material hazards for infrastructure - floods, storms, wildfires, and heat stress - which collectively account for the large majority of climate-related economic losses globally. Hail coverage will be assessed as part of our ongoing indicator development roadmap.

Q How are tipping points and feedback loops reflected in the hothouse scenarios?

The current ClimateMetrics methodology does not explicitly model non-linear climate tipping points (e.g. permafrost collapse, ice sheet instability, Amazon dieback) or feedback loops. The physical hazard projections are based on mainstream IPCC-consistent climate models (RCP 4.5 and RCP 8.5), which capture gradual physical intensification but do not incorporate abrupt or cascading climate system transitions.

This is a known limitation of the current state of the art in operational climate risk tools. Tipping-point dynamics remain an active area of climate science research and are not yet routinely embedded in the scenario frameworks used for financial risk assessment - including NGFS scenarios. We disclose this limitation explicitly in our methodology documentation. For scenarios involving non-linear system dynamics, we recommend complementing ClimateMetrics outputs with qualitative scenario analysis.

Q Are the assumptions about tipping points, compound climate events, and systemic risks available to investors so that model risk is transparent?

Yes - we are committed to full methodological transparency. The complete methodology is publicly available in our Methodological Note, which documents all data sources, scenario assumptions, damage functions, and scoring logic. This includes explicit disclosure of limitations such as: the absence of tipping-point modelling, the use of sector-level proxy data where asset-specific data is unavailable, country proxy damage functions where calibrated curves do not exist, and the capping of tropical cyclone projections at 2060.

We also provide a Data Quality Score (DQS) alongside each rating, which quantifies the reliability of the underlying input data (geolocation precision, emissions data quality), helping investors understand the confidence level associated with each assessment. Model risk is an important concept and transparency is the only credible response to it.

4. Transition Risk

Q Can you explain how transition risk is calculated at the asset level?

Transition risk is split into two sub-risks, each producing its own A-G rating.

Policy & Technology risk measures exposure to carbon pricing. We estimate the asset's Scope 1+2 carbon intensity per revenue (tCO₂e per M\$), project it forward using country- and sector-level decarbonisation pathways, and multiply it by the country-specific carbon price trajectory (sourced from Oxford Economics). This produces a forward-looking carbon cost burden relative to revenue, which is log-transformed and scored against the percentile distribution of our infrastructure universe.

Reputation & Market Preferences risk captures exposure to shifts in revenue dynamics driven by the climate transition. This is modelled through a revenue indicator: the asset's projected revenue trajectory under a given climate scenario is compared against a reference revenue path that evolves only with inflation. The divergence between these two paths drives the Reputation & Market rating.

The two sub-risk scores are then weighted - using the sector-specific Scope 3-to-Scope 1+2 ratio as a weighting factor - and combined into the overall Transition Exposure Rating (TER, A-G).

Q Why is the Reputation & Market demand dimension of transition risk becoming increasingly important?

Carbon pricing is the most legible transition risk because it has a direct, quantifiable cost channel. But the assumption that carbon pricing is the dominant transition risk vector is increasingly challenged by market evidence: changes in revenue trajectories driven by shifts in consumer demand, investor preferences, and competitive dynamics are often faster and less predictable than regulatory carbon price signals.

For infrastructure assets specifically, the commercial viability of certain asset types is being reshaped by the transition - not primarily because of a carbon tax, but because the revenue environment is changing structurally. An airport facing a structural decline in short-haul demand, or a gas distribution network whose throughput assumptions are challenged by electrification, faces a transition risk that a pure carbon cost model would materially understate.

Q Can users add their own assumptions, such as custom carbon price expectations?

In the standard licence, users can choose from four transition scenario trajectories (Baseline, Conservative, Expected, Progressive), which embed different carbon price pathways sourced from Oxford Economics. Fully custom carbon price trajectories are not available in the standard plan, but the climateMetrics Pro plan supports custom scenarios tailored to client-specific assumptions.

Users can also supply their own reported Scope 1 and 2 emissions, asset revenues and values, and custom resilience strategy effectiveness values - each of which meaningfully improves rating precision.

Q How is Scope 3 handled?

Scope 3 emissions are not directly input by users but are estimated by the platform using a sector- and country-level input-output model. The Scope 3 intensity is derived as a ratio of the asset's Scope 1+2 intensity, calibrated at the TICCIS superclass level. This ratio is further split into upstream Scope 3 (supply chain) and downstream Scope 3 (product use), allowing separate upstream and downstream Reputation & Market Preferences ratings to be computed.

The resulting Scope 3 carbon cost indicator (Scope 3 intensity x country carbon price) contributes to the Transition Exposure Rating alongside the Scope 1+2 carbon cost. The relative weight of Scope 3 vs. Scope 1+2 in the TER reflects the sector's typical S3-to-S1+2 ratio. Upstream and downstream Scope 3 intensities are also displayed in the dashboard as contextual indicators.

Q What GHG data does the platform use, and where does it come from?

For Scope 1 and 2 emissions, users can input their own reported figures (in tCO₂e per year). If not provided, the platform estimates them using TICCIS sector-level default emissions intensities per revenue, sourced from Scientific Climate Ratings' infrastructure universe of ~6,000 companies.

For Scope 3 emissions, the platform estimates upstream and downstream intensities using an input-output model calibrated at sector and country level, applied as a ratio to the Scope 1+2 intensity. All emissions data feeds into the Transition Exposure Rating (TER). Providing actual reported emissions improves precision and is reflected in the Data Quality Score (DQS) associated with the rating.

Q How is the impact of climate change on the supply chain of an asset reflected in the tool?

Supply chain climate risk is partially captured through the Scope 3 component of the Transition Exposure Rating. We estimate upstream and downstream Scope 3 emissions intensities using sector- and country-specific input-output ratios applied to the asset's Scope 1+2 intensity. These are then combined with carbon price projections to produce upstream and downstream carbon cost indicators, which feed into the Reputation & Market Preferences sub-risk.

Physical supply chain risks - such as a supplier's facility being flooded - are not currently modelled. The platform assesses each asset individually based on its own footprint and does not propagate physical damage through supply chain networks. For assets with critical supply chain dependencies, we recommend supplementing the platform's outputs with operational risk analysis.

5. Quantification in financial impacts

Q How do peer-reviewed damage functions translate climate hazards into financial impacts?

Damage functions are the mathematical bridge between a physical hazard intensity - say, a flood depth of 1.5 metres - and its financial consequence for a specific asset type. The function maps hazard intensity to a damage fraction [0-1], representing the share of the asset's replacement value that would be destroyed at that intensity level.

This relationship is asset-type- and country-specific. ClimateMetrics implements 32 distinct flood damage curves (by country x TICCIS sector subclass) and dedicated wind damage curves calibrated to infrastructure building typologies, all derived from peer-reviewed academic sources.

The output investors receive is an Expected Damage (ED) - the integral of the damage curve across all return periods, weighted by their annual exceedance probabilities. This yields an annualised damage fraction directly comparable across assets and directly usable in NAV analysis, stress-testing, and insurance premium benchmarking.

Q Can you walk through how physical exposure is translated into financial impact, step by step?

The process follows five steps:

1. Hazard mapping: we extract the hazard intensity at the asset's precise location for each return period (e.g. flood depth in metres at a 100-year return period).
2. Damage functions: we apply sector- and country-specific damage functions that map hazard intensity to a damage fraction (0-100% of asset replacement value).
3. Expected Damage integration: we compute the probability-weighted average damage across all return periods, yielding an Annualised Average Expected Loss (AAEL) expressed as a % of asset value per year.
4. Revenue disruption: for applicable hazards (particularly heat stress and disruption from floods/storms), revenue impact is also computed using sector-specific disruption factors.
5. Rating: the financial damage metric is scored against a reference distribution of ~6,000 infrastructure assets to produce the A-G Physical Exposure Rating (PER).

This approach means outputs are directly usable for financial modelling - adjusting NAV assumptions, sizing insurance coverage, or stress-testing cash flows.

6. Ratings and benchmarking

Q How does the rating scale work, and what does an A-G rating mean?

Ratings from A (best) to G (worst) are derived by converting continuous exposure scores - ranging from 1 to 100 - into letter categories following an idealised, near-normal distribution. The score reflects an asset's percentile rank within our infrastructure universe of approximately 6,000 assets, calibrated at the Expected scenario and 2035 time horizon.

The distribution is designed to be symmetrically spread across seven categories: A (scores 1-5), B (5-17.5), C (17.5-37.5), D (37.5-62.5), E (62.5-82.5), F (82.5-95), and G (95-100). This consistent anchoring ensures that a rating of 'C' carries the same meaning today as it will in three years, and across different asset types and geographies.

Q Do you consider the PER and TER independently static, or dynamically weighted depending on sectoral exposure and financing horizon?

The relative weights of PER and TER in the CER are not static - they are derived from the estimated financial impact of physical vs. transition risks on Net Asset Value at the geo-sector level (TICCS sector x climate zone). This means the weighting varies by sector and geography rather than being a fixed split. For example, a coastal power plant in a tropical region will naturally carry higher physical weight than a gas network in a temperate country.

The weighting does not currently vary by financing horizon in the standard platform, though the Pro plan allows custom time horizon configurations that implicitly affect the relative risk balance. As climate risk integration in debt pricing, insurance access, and blended finance structures matures, we expect ratings like the CER to increasingly inform those instruments - though this integration is driven by market adoption rather than a product feature we control directly.

Q Do the approximately 6,000 assets in the reference universe reflect public or private market data?

The reference distribution used to calibrate ClimateMetrics' ratings is drawn from the Unlisted Infrastructure Universe maintained by Scientific Infra & Private Assets (SIPA). This universe covers approximately 6,000 to 9,100 privately-owned infrastructure companies across 27 active national markets, all classifiable under TICCS.

This is a key differentiator: our ratings are benchmarked against a private infrastructure universe, not against public equities or real estate indices. This makes the benchmarking directly relevant to the institutional infrastructure investment community.

Q How are portfolio ratings calculated?

Portfolio-level ratings are computed as arithmetic averages of asset-level numeric ratings. Letter ratings (A-G) are mapped to numeric values (A=1 through G=7), averaged across all assets with valid ratings, and the resulting mean is rounded to the nearest integer and mapped back to a letter. This is computed separately for the CER, PER, and TER. One portfolio aggregation counts as one request.

Q How is benchmarking against peer assets used, and why does it matter?

Context is often more decision-relevant than an absolute number. Knowing that an asset has a 2.3% annualised expected flood damage is informative; knowing that this places it in the top quartile of exposure relative to comparable infrastructure assets is actionable.

ClimateMetrics ratings are anchored to a fixed reference distribution of approximately 6,000 private infrastructure assets across 8 TICCS superclasses. This means an 'E' rating carries a consistent meaning across asset types, time, and geographies - not a relative ranking within a single client's portfolio. This consistency is particularly valuable for regulatory disclosure, transaction due diligence, and LP-level climate reporting.

7. Resilience & Decarbonisation

Q Does ClimateMetrics account for adaptation measures?

Yes. Adaptation and resilience are integrated through our proprietary ClimaTECH database, developed by EDHEC Climate Institute. ClimaTECH covers more than 100 distinct resilience and decarbonisation strategies across 101 infrastructure sectors (approximately 1,800 sector-strategy applications). Each strategy's effectiveness is assessed from over 200 academic papers and validated by an expert review board.

For physical risk, users can declare specific resilience measures - flood defences, fire-resistant materials, structural hardening, etc. The platform then adjusts the damage estimates accordingly to produce a Post-Resilience CER (PCER). For decarbonisation, users can declare technologies that reduce Scope 1, 2, or 3 emissions, expressed as a percentage reduction applied to the baseline emissions intensity.

If the exact effectiveness of a measure is unknown, ClimaTECH's default value is applied. Users can also override this with a custom effectiveness percentage if they have asset-specific data.

Q How are the effectiveness percentages for resilience measures calculated?

The effectiveness percentages are sourced from the ClimaTECH database, which quantifies how much each strategy reduces expected damage at each return period (e.g. 20-year, 100-year, 500-year events). The platform applies these reductions to the hazard-specific damage curves.

The underlying logic is conservative: a strategy rated at 30% damage reduction for a 100-year flood is assumed to also protect against milder events, but provides no protection for events more severe than the declared protection level. The resulting adjusted damage feeds into the recalculated PCER.

Q Is the resilience measure input qualitative, and is the exposure reduction calculated automatically?

The input process is structured but straightforward: users select the relevant resilience measures from a questionnaire in the platform (e.g. flood barrier, fire-resistant cladding, heat insulation), specify the protection level or effectiveness, and the platform automatically computes the adjusted damage curve using the ClimaTECH database. The result - the Post-Resilience CER (PCER) - is generated without requiring case-by-case analyst support.

For the standard licence, all adjustments are fully automated based on ClimaTECH default effectiveness values. The Pro plan includes advisory support for more complex resilience modelling, such as bespoke engineering solutions or multi-measure portfolios where user-defined effectiveness parameters are appropriate.

Q How do you account for company-specific adaptation capabilities beyond sector-level averages?

Yes - company-specific adaptation and transition measures can be declared through the questionnaire in the platform. Users can input resilience strategies and decarbonisation strategies that override the sector-level defaults. Effectiveness values are drawn from the ClimaTECH database and can also be customised by the user if asset-specific data is available.

This mechanism allows assets with genuinely superior resilience or decarbonisation trajectories to receive adjusted ratings (the PCER) that reflect their actual profile rather than a sector average. Declared strategies are subject to review by our team to ensure credibility of the declared effectiveness.

Q Where do decarbonisation strategies come from, and how are they validated?

Decarbonisation strategies are declared by the asset owner or operator through a questionnaire in the platform. Scientific Climate Ratings then reviews and validates each strategy using the ClimaTECH database, which assesses each strategy's effectiveness in terms of avoided emissions per scope. Companies must specify which share of emissions the technology impacts, and may be asked to provide supporting documentation.

If a company does not provide this information, no adjustment is made - the conservative baseline (unmitigated emissions intensity) is retained. This design ensures that the adjusted rating rewards companies that can substantiate their measures, without penalising those that have not yet engaged with the questionnaire.

Q Is the quality of the adaptation and resilience assessment dependent on accurate inputs from companies?

Yes - and this is an important nuance. The output is only as reliable as the declared strategies. ClimaTECH's effectiveness values represent the best available peer-reviewed evidence at the sector and hazard level, but the strategies themselves are self-declared by the company. SCR reviews each response for plausibility and requests supporting materials where appropriate. For users who cannot provide detailed strategy information, the model defaults to the unmitigated baseline, which is a conservative and defensible position for reporting purposes.

8. Use cases

Q How can climate risk ratings be integrated into broader risk management or investment decision frameworks?

ClimateMetrics is designed to sit alongside existing investment and risk workflows. In practice, users integrate the outputs in several ways: as an additional screening or scoring layer at the asset selection or due diligence stage; as an input to internal models (e.g. adjusting discount rates or cash flow assumptions); as a portfolio monitoring metric updated annually; and as structured data exports (Excel, PDF) that feed directly into TCFD, CSRD, ISSB, and SFDR disclosures.

The climateMetrics Pro plan includes direct API access, enabling automated integration with portfolio management systems, risk platforms, and proprietary internal tools.

Q Can the platform support operational planning decisions such as offshore operations or weather-window management?

The platform is primarily focused on assessing and quantifying the strategic, long-term financial impacts of climate risks - not supporting day-to-day operational planning such as offshore operations management or weather-window optimisation. Long-term risks relevant to these operational applications could potentially be informed by some of the underlying climate data, but this is not the primary scope of the framework.

Q How does ClimateMetrics help with insurance and insurability?

ClimateMetrics can meaningfully support insurance-related applications. The Annualised Average Expected Loss (AAEL) metric - expressed as a percentage of asset replacement value per year - is structurally equivalent to the pure risk premium concept used in insurance pricing. This gives underwriters a forward-looking, scenario-consistent loss estimate to inform premium setting and risk transfer decisions.

Second, the platform's physical risk output, broken down by hazard and return period, can help identify assets that may be becoming difficult to insure as climate risk intensifies - particularly those with high expected losses at short return periods.

ClimateMetrics does not set insurance premiums or define insurability thresholds - these are commercial decisions made by insurers based on their own underwriting criteria. Our role is to provide the rigorous, asset-specific climate risk data that feeds into that process.

Q Is there a feature to help investors assess nature-related assets such as carbon and biodiversity credits?

Nature-related assets such as carbon credits and biodiversity credits are not currently within the scope of ClimateMetrics, which is focused on physical infrastructure and built assets. The platform does not provide tools for valuing or stress-testing nature-based financial instruments.

For investors seeking to assess nature-related risks, we would suggest looking at frameworks such as TNFD (Taskforce on Nature-related Financial Disclosures) and dedicated biodiversity risk tools. These are complementary to, but distinct from, the climate risk quantification that ClimateMetrics provides.

Q Can the tool be used to forecast the climate financial impact of anthropogenic activities such as deforestation?

No - ClimateMetrics is designed to assess the climate risk exposure of existing physical assets, not to model the climate impacts of human activities such as deforestation. The platform measures how climate change affects the value and operations of an asset, rather than the reverse.

That said, deforestation and land-use change indirectly affect some ClimateMetrics outputs: land-use change influences wildfire risk and local heat dynamics, which are embedded in the underlying climate datasets. But the platform does not provide a tool for quantifying the climate consequences of a specific land-use decision.

Q What kind of support and training do you offer?

All licences include access to a support hotline (chatbot and working-hours telephone), a one-hour onboarding session to ensure correct setup, three dedicated training sessions, and access to training materials including the user guide and walkthrough videos.

The climateMetrics Pro plan additionally includes access to a dedicated Customer Success Manager and tailored training sessions on request, as well as advisory support on resilience and decarbonisation strategy modelling.

Q What practical actions can asset managers take based on ClimateMetrics ratings?

ClimateMetrics outputs are designed to enable three broad categories of action:

- Risk identification and prioritisation: identifying which assets carry the highest climate exposure and which hazards are most material - enabling targeted due diligence and capital allocation.
- Resilience planning: using the PCER to model the impact of specific adaptation investments on ratings, and to build a business case for resilience capital expenditure.
- Regulatory disclosure: structuring climate risk data in the formats required by TCFD, CSRD/ESRS E1, ISSB/IFRS S2, and SFDR.

Q Why do investors still struggle to translate climate science into investment-relevant financial metrics, and how does ClimateMetrics address that?

The gap is structural. Climate science produces outputs - temperature pathways, hazard probabilities, scenario trajectories - that speak a fundamentally different language from the metrics that drive investment decisions: expected losses, NAV impacts, and revenue-at-risk. Bridging that gap requires translating hazard intensity into asset-specific damage through sector- and country-calibrated functions, expressing those damages as a fraction of asset value or revenue, and doing so consistently across thousands of assets so results are comparable, not merely informative.

A second barrier is trust. The market has been saturated with qualitative ESG scores and opaque ratings that are difficult to audit or defend internally. Finance teams rightly push back on outputs they cannot trace back to assumptions.

ClimateMetrics was built to close that gap - by grounding every output in quantitative, peer-reviewed science, maintaining full methodological transparency, and expressing results in financial terms usable by a CFO or risk committee. Outputs are not 'high/medium/low' flags - they are annualised expected damage fractions, revenue impacts, and A-G ratings anchored to a reference distribution of approximately 6,000 infrastructure assets.

Q Why was it important to integrate physical and transition risk into a single framework, and are clients more focused on one versus the other?

Physical and transition risks are not independent - they are driven by the same underlying dynamic: the trajectory of climate change. A world that warms aggressively (high physical risk) is also a world where decarbonisation efforts intensify and carbon prices rise (high transition risk). The Climate Exposure Rating (CER) captures this co-movement by weighting physical and transition exposure based on their relative financial impact on Net Asset Value (NAV) at the geo-sector level.

Assessing either dimension in isolation can be materially misleading. A renewable energy asset may appear exposed to transition risk metrics in isolation, but relatively resilient once physical exposure is correctly factored in. Conversely, a coastal fossil fuel plant carries compounding risks from both dimensions.

In terms of client focus today, there is a clear tilt toward physical risk, driven by regulatory requirements - TCFD, CSRD/ESRS E1, and ISSB/IFRS S2 all mandate physical risk disclosure. Transition risk attracts more interest among clients with large carbon-intensive portfolios or strong ESG mandates.

Q What are the consequences of failing to comply with IFRS S2 (ISSB) and/or CSRD?

Both IFRS S2 and CSRD (ESRS E1) require detailed, auditable, and scenario-consistent disclosures on physical and transition climate risks, including asset-level climate exposure and GHG emissions across all scopes. The specific penalties for non-compliance depend on jurisdiction and entity type, but commonly cited consequences include: financial sanctions from national competent authorities; reputational damage and restatement risk; exclusion from capital markets or financing programmes with ESG conditions; and increasing investor scrutiny.

ClimateMetrics directly supports compliance by providing the granular, auditable, peer-reviewed data required for these disclosures. The platform's Excel and PDF exports are structured to map to the key disclosure requirements of TCFD, CSRD/ESRS, ISSB/IFRS S2, and SFDR.

Q Where do you see climate risk analysis evolving over the next five years, and what will investors expect that they do not yet receive today?

Broader hazard coverage. Drought, earthquake, subsidence, and sea-level rise are increasingly relevant for agriculture, coastal assets, and geologically active regions. These are already on our near-term development roadmap.

Deeper financial integration. The field is moving from exposure metrics to full financial risk integration - connecting climate damage to cash flow models, discount rates, and asset valuations. Metrics like Probable Maximum Loss and Cost of Inaction are coming into focus.

Higher regulatory standards. The convergence of IFRS S2, CSRD, SFDR, and central bank stress-testing frameworks will raise the technical bar for what is accepted. Generic qualitative assessments will not satisfy regulators.

Dynamic, API-driven analysis. The market will increasingly expect near-real-time capabilities rather than annual static snapshots. Integration with portfolio management systems via API, and the ability to model the financial impact of specific adaptation investments on a rating, are capabilities clients will treat as table stakes within the next few years.

Q With Article 6 and sovereign carbon strategies accelerating, do you see future climate ratings integrating carbon credit generation capacity as part of resilience or transition scoring?

This is an important forward-looking question. Today, ClimateMetrics does not incorporate carbon credit generation capacity as a component of resilience or transition scoring. The TER is based on carbon cost exposure (emissions intensity x carbon price) and revenue trajectory shifts - not on offsetting mechanisms.

As Article 6 markets mature and sovereign carbon strategies become more quantifiable, we see merit in exploring how verified carbon removal or avoidance capacity could be treated as a risk-reducing factor in transition scoring. This would require robust, auditable data on carbon credit quality and permanence - a challenge the market is still working through. We will monitor developments in this space closely.

9. Pricing & Commercial

Q What is the pricing of ClimateMetrics?

Pricing is based on Assets Under Management (AuM) or equivalent, and is fully transparent. The standard ClimateMetrics annual licence ranges from EUR 30,000 per year (AuM below EUR 0.5bn) to EUR 90,000 per year (AuM above EUR 100bn). The reference base price - for AuM between EUR 10bn and EUR 25bn - is EUR 60,000 per year. A 10% discount applies on multi-year contracts.

The standard licence includes 5,000 asset requests per year, unlimited users, full access to all hazard modules, all scenarios and time horizons, and the complete output suite (Excel and PDF exports). The climateMetrics Pro plan - which adds custom scenarios, API access, unlimited requests, and a dedicated Customer Success Manager - is priced on a case-by-case basis.